

Physics-Guided Allocation of Conductive Particles in Granular Beds: A Hybrid DEM, Thermal-Network and Graph-Learning Framework

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Abstract

The effective thermal conductivity of a granular bed is governed not only by porosity, particle-size distribution and constituent conductivities, but also by the topology of the particle contact network. In a binary bed containing a limited fraction of highly conductive particles, the same material fraction can therefore produce different heat-transfer performance depending on where those particles are placed. Existing thermal discrete-element and resistance- network models mainly evaluate the forward relation from an already specified packing and material distribution to heat transfer. Conversely, graph-learning and inverse-design studies have generally focused on continuous or architected materials rather than discrete material allocation on mechanically realizable granular contact networks. This study proposes a hybrid framework in which the discrete-element method (DEM) generates physically admissible packings, a steady thermal resistor network provides exact and inexpensive physics labels, and graph-based optimization identifies which particles should receive a high-conductivity material under a fixed material budget. The thermal network contains both solid-contact conduction and conduction through a stagnant interstitial fluid; no imposed fluid motion or convective transport is assumed. Random allocation, degree and centrality rankings, continuous-relaxation methods and greedy physics-based selection provide mandatory baselines. A graph neural network (GNN) is introduced only if it can learn a transferable placement policy that generalizes to unseen DEM packings or larger systems. The central hypothesis is that conductivity enhancement depends on the formation of boundary-spanning, low-resistance pathways and cannot be determined reliably from porosity, particle-size distribution or isolated node scores alone. An initial pilot uses ten independent 500-particle packings with identical number-based particle-size distribution and

nominal porosity. The intended outcome is a reproducible physics-guided design rule for allocating a limited conductive phase in granular thermal-storage or heat-transfer media, together with quantitative evidence of when learned graph policies add value beyond conventional network heuristics.

Keywords: granular beds; discrete-element method; thermal resistor network; effective thermal conductivity; material allocation; graph neural network; inverse design; hybrid modelling

1 Motivation, research gap and hypothesis

Heat conduction through granular media is controlled by particle properties, contact areas, packing density, interstitial phases and the organization of the contact network. Classical studies and more recent thermal DEM formulations have established physically meaningful forward models for conduction through particle assemblies [1, 2, 3]. These models can resolve packing heterogeneity more directly than correlations based only on bulk porosity or coordination number. They do not, however, automatically answer an inverse design question: given a mechanically fixed packing and a limited quantity of an expensive conductive material, which particles should be made conductive to maximize heat transfer between two boundaries?

This distinction is important. If the fraction of conductive particles is fixed, random mixing may waste material in isolated clusters, transverse branches or redundant parts of an already conductive pathway. Selecting the highest-degree particles is also not necessarily optimal, because a locally well-connected node can be poorly positioned relative to the hot and cold boundaries. The useful arrangement must account simultaneously for local connectivity, particle size, contact conductance, direction relative to the temperature gradient, and the global formation of competing or complementary paths through the bed.

Graph-based inverse design has been demonstrated for architected lattice materials, including differentiable message-passing approaches that modify local graph attributes to achieve target properties [5]. The proposed study is different in both physical system and design variable. The graph is not an arbitrary lattice to be generated or geometrically deformed; it is a contact network produced by DEM at prescribed porosity and particle-size distribution. Its geometry remains fixed, and the inverse problem is a binary, fixed-

budget assignment of particle material. The scientific question is therefore whether a small conductive phase can be allocated more effectively by exploiting the actual topology of a realizable granular packing.

The central hypothesis is:

For granular beds with identical porosity, particle-size distribution and conductive-particle fraction, effective conductivity depends strongly on the global arrangement of the conductive phase. Boundary-aware, cooperative selection of particles can create low-resistance spanning pathways that cannot be identified reliably by random allocation or isolated node-centrality scores.

This hypothesis is falsifiable. If optimized assignments do not consistently outperform random and simple heuristic allocations across unseen packings, the placement problem is not practically important in the investigated regime. If a deterministic physics-based algorithm performs as well as a GNN, the graph learning component is unnecessary and should not be presented as the main contribution.

2 Hybrid modelling concept and novelty

The framework separates three roles. DEM generates mechanically realizable particle positions, sizes and contacts. A thermal-network solver evaluates any material assignment by enforcing energy conservation. Optimization or graph learning searches the discrete assignment space without replacing the physical thermal model.

For a packing graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, let $x_i \in \{0, 1\}$ indicate whether particle i is assigned the high-conductivity material. The fixed material budget is

$$\sum_{i \in \mathcal{V}} x_i = M, \quad (1)$$

where M is the prescribed number of conductive particles. Particle conductivity is then

$$k_i = k_L + x_i (k_H - k_L). \quad (2)$$

For every internal particle, steady energy conservation requires

$$\sum_{j \in \mathcal{N}_i} G_{ij}(T_j - T_i) = 0, \quad (3)$$

where G_{ij} is the total solid-contact and stagnant-fluid conductance between neighbouring particles i and j . After applying hot and cold boundary temperatures, the resulting heat rate Q gives

$$k_{\text{eff}} = \frac{QH}{A(T_h - T_c)}. \quad (4)$$

The inverse problem is

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \{0,1\}^N} k_{\text{eff}}(\mathcal{G}, \mathbf{x}) \quad \text{subject to Eq. (1)}. \quad (5)$$

The main novelty is not the separate use of DEM, network conduction or a GNN. It is the controlled inverse allocation of a fixed conductive-particle budget directly on multiple physically realizable DEM contact networks, with the thermal equations retained as the evaluator. The study will quantify which topological mechanisms make an allocation effective, whether optimal patterns transfer across stochastic packings, and whether graph learning offers a reproducible advantage over transparent physics and network baselines. This is more specific and defensible than claiming the first use of machine learning for granular thermal conductivity.

Figure 1 illustrates the physical problem. The particle geometry and contact topology are fixed after DEM generation. Only the binary material labels are changed. Heat can pass through direct solid contacts and through the stationary fluid occupying the void space. The latter contributes thermal conduction without requiring fluid motion or CFD.

3 Proposed work packages

WP1: Controlled DEM packing ensemble

LAMMPS is used to generate independent random packings while keeping the number-based particle-size distribution and nominal porosity identical. The current pilot contains ten accepted packings. Each packing has 500 spherical particles divided into 100 particles of diameter 1.5 mm, 300 particles of diameter 2.0 mm and 100 particles of diameter

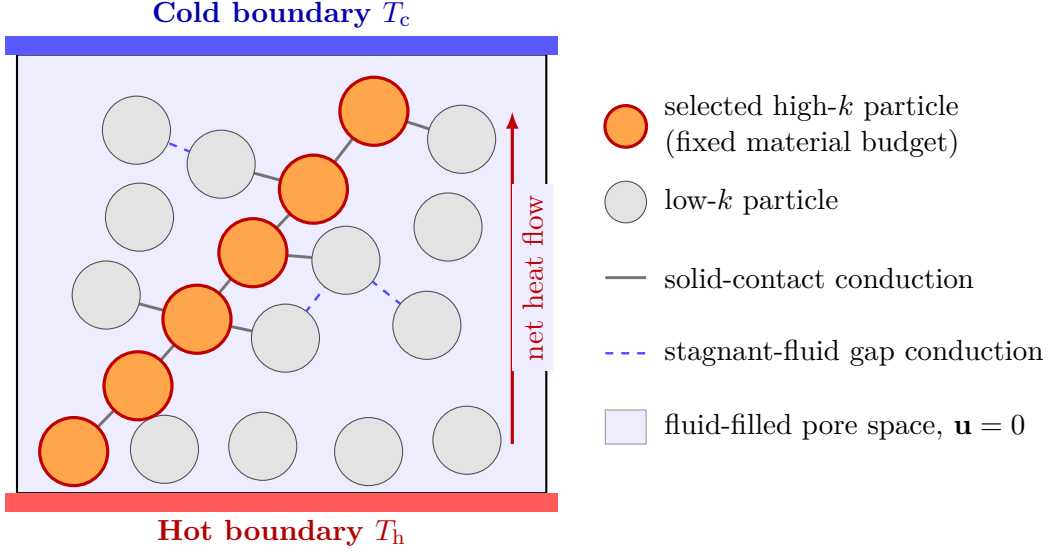


Figure 1: Schematic of the inverse allocation problem. DEM fixes the packed-bed geometry. A prescribed number of particles receives the high-conductivity material. The steady thermal network combines conduction through particle contacts and through the stagnant fluid in neighbouring gaps.

2.5 mm. The nominal porosity is 0.38. Particle centres are initialized randomly at reduced size and the three diameter classes are expanded quasistatically inside a fixed-volume cubic cell. Changing only the random seed produces different contact topologies without changing the prescribed bulk descriptors.

For each realization, the final particle and contact files are retained. Acceptance checks include exact particle counts, reconstructed particle-size distribution, nominal porosity, kinetic energy, pressure, normalized overlap, coordination statistics, connected-component size and hot-to-cold spanning connectivity. The ten cases are sufficient to develop and debug the complete workflow, but they are not automatically sufficient for the final statistical or GNN study. Dataset expansion is decided from the observed between-packing variability and learning curves in WP4.

Pilot component	Current specification
Independent realizations	10 accepted DEM packings
Particles per realization	500
Number-based PSD	100/300/100 particles at 1.5/2.0/2.5 mm
Nominal porosity	0.38 in a fixed cubic cell
Varying quantity	random seed and resulting contact topology
Fixed quantities	particle counts, PSD, porosity, material density and DEM settings
Primary outputs	particle coordinates/radii, particle contacts, wall contacts and packing summary

WP2: Physical thermal-network solver

Every DEM packing is converted to a weighted graph. Nodes represent particles, particle–particle contacts define solid-conduction edges, and wall contacts couple the graph to hot and cold reservoirs. The void space contains a stagnant interstitial fluid. Its velocity is assumed to be zero, $\mathbf{u} = 0$, so it transports energy by conduction but not by forced or natural convection. The steady fluid energy equation therefore reduces to

$$\nabla \cdot (k_f \nabla T) = 0. \quad (6)$$

Rather than resolving this equation by CFD, a provisional reduced model connects nearby non-contacting particles by fluid-gap conductances derived from their radii, surface separation, fluid conductivity and an effective exchange area. The total pair conductance is

$$G_{ij}^{\text{total}} = G_{ij}^{\text{contact}} + G_{ij}^{\text{gap}}. \quad (7)$$

In the present implementation, contacting particles receive the solid-contact term, whereas selected nearby non-contacting pairs receive the stagnant-fluid gap term. This pairwise gap construction must pass parameter-sensitivity checks before it can be used for physical conclusions. Radiation is initially neglected and can later be activated for high-temperature

applications.

A contact constriction model provides a suitable first conductance law. With contact radius a_{ij} , the resistance of a contact between particles i and j can be written as

$$G_{ij} = \left(\frac{1}{2k_i a_{ij}} + \frac{1}{2k_j a_{ij}} \right)^{-1} = \frac{2a_{ij}k_i k_j}{k_i + k_j}. \quad (8)$$

For Hertzian geometry, $a_{ij} = \sqrt{R_{ij}^* \delta_{ij}}$, where $R_{ij}^* = R_i R_j / (R_i + R_j)$ and δ_{ij} is overlap. Because DEM overlap depends on stiffness and confining state, this choice requires a sensitivity study. A normalized contact-area model, or a prescribed interfacial resistance, will be used as a comparison to ensure that the allocation conclusions are not an artefact of numerically softened particles.

The stagnant-fluid approximation is checked using a pore-scale Rayleigh number,

$$Ra_p = \frac{g\beta\Delta T d_p^3}{\nu\alpha}. \quad (9)$$

For small pores and moderate temperature differences, a sufficiently small Ra_p supports neglecting buoyancy-driven circulation. The scope is therefore a static packed bed, not a bed undergoing forced through-flow. If an application contains appreciable imposed flow, convection would require a different model and is outside the first study.

The sparse linear system from Eq. (3) is solved for all particle temperatures. Verification begins with homogeneous beds ($k_i = k_L$ for all particles). Required checks include positive conductances, temperatures bounded by the wall values, equality of hot- and cold-side heat rates to solver tolerance, invariance to a common scaling of temperature difference and reproducibility under node reordering.

Numerical verification on the first pilot packing

The contact-only solver has been numerically verified on realization `packing_seed_18427`. The DEM graph contains 500 particles and 1343 particle–particle contacts, corresponding to a mean coordination number $2N_c/N = 5.372$. One boundary-spanning component contains 497 particles. Three individual particles have no solid-contact path to either thermal wall. In the contact-only limit they carry no heat and their absolute steady temperature is undefined; they are therefore excluded from the linear system and reported

explicitly rather than assigned an artificial temperature. The solved matrix has size 497×497 with 3183 nonzero entries. The bottom and top boundaries are connected through 42 and 41 particle–wall contacts, respectively.

Table 1 summarizes three calculations. The baseline uses homogeneous particles with $k_s = 1 \text{ W m}^{-1} \text{ K}^{-1}$ and a temperature difference of 1 K. Doubling all particle conductivities doubles both the total heat rate and effective conductivity while leaving the normalized temperature field unchanged. Increasing the imposed temperature difference by a factor of ten increases the heat rate by the same factor while leaving k_{eff} unchanged. These are required consequences of the linear steady conduction equations.

Table 1: Numerical verification of the contact-only thermal-network solver for the first DEM realization.

Calculation	k_s [W/(m K)]	ΔT [K]	Q_h [W]	k_{eff} [W/(m K)]
Baseline	1.0	1	$1.44369666 \times 10^{-3}$	0.0939256384
Conductivity scaling	2.0	1	$2.88739332 \times 10^{-3}$	0.1878512768
Temperature scaling	1.0	10	$1.44369666 \times 10^{-2}$	0.0939256384

The hot–cold relative energy-balance errors are 9.54×10^{-14} , 9.54×10^{-14} and 9.13×10^{-15} for the three calculations. The numerical scaling and conservation tests therefore pass to approximately machine precision. This establishes numerical verification of the assembly and solution procedure. It is not yet a physical validation of the predicted conductivity magnitude, which still requires sensitivity to the overlap-based contact area, stagnant-fluid model and, ultimately, comparison with an analytical, literature or experimental reference.

Homogeneous ensemble baseline and fluid-model sensitivity

The homogeneous contact-only calculation was repeated for all ten accepted packings. Every realization contains one hot-to-cold spanning component. The mean contact count is 1348.3 ± 16.5 , and the mean coordination number is 5.393 ± 0.066 ; both have a coefficient of variation of 1.22%. The resulting effective conductivity is

$$k_{\text{eff}} = 0.09329 \pm 0.00351 \text{ W m}^{-1} \text{ K}^{-1}, \quad (10)$$

with a range of $0.08615\text{--}0.09717\text{ W m}^{-1}\text{ K}^{-1}$ and a coefficient of variation of 3.77%. Between one and eleven particles per realization are thermally unanchored in the contact-only limit, while the largest spanning component contains 489–498 particles. The maximum relative hot–cold energy- balance error is 1.31×10^{-12} . These results establish a reproducible stochastic packing baseline and show that heat transfer varies more strongly than contact count alone.

A stagnant-air sensitivity test was then performed for realization `packing_seed_18427` using $k_f = 0.026\text{ W m}^{-1}\text{ K}^{-1}$. With a regularization of $0.001d_{\text{ref}}$, increasing the maximum fluid-gap distance from $0.05d_{\text{ref}}$ to $0.10d_{\text{ref}}$ and $0.20d_{\text{ref}}$ increased the number of fluid edges from 245 to 427 and 784, respectively. The corresponding effective conductivities were 0.10578, 0.11825 and 0.14498 $\text{W m}^{-1}\text{ K}^{-1}$. By comparison, changing the gap regularization from $0.0005d_{\text{ref}}$ to $0.002d_{\text{ref}}$ at the fixed cutoff $0.10d_{\text{ref}}$ changed k_{eff} only from 0.11885 to 0.11717 $\text{W m}^{-1}\text{ K}^{-1}$.

The numerical solutions remain conservative and all 500 particles become thermally anchored when the fluid edges are enabled. However, the large cutoff sensitivity shows that the current pairwise fluid-gap construction is not yet a validated physical model: the cutoff controls both how many pairs are linked and the assumed exchange area. Consequently, the verified contact-only network is retained for the first fixed-budget allocation pilot. Physical conclusions including the stagnant fluid will require a geometry-defined pore-network model, preferably based on a weighted Voronoi or radical-Delaunay description for the polydisperse packing.

WP3: Fixed-budget allocation and transparent baselines

The high-conductivity fraction and conductivity contrast are varied only after the homogeneous solver is verified. Initial conductive fractions of 5, 10, 15 and 20% and several values of k_H/k_L can reveal when placement matters. For $N = 500$, the 10% case corresponds to selecting exactly $M = 50$ particles. Thousands of assignments can be evaluated on each fixed packing without rerunning DEM.

The proposed hierarchy of allocation strategies is shown in Table 2. Random allocation is not a single result: it must be sampled repeatedly to obtain a distribution. Degree ranking tests whether local connectivity alone is sufficient. Heat-path ranking and fixed-budget swap optimization test whether the temperature-gradient direction and cooperative

changes in network resistance are essential.

Table 2: Mandatory allocation baselines and the questions they test.

Method	Allocation principle	Purpose
Random	uniform selection of exactly M particles; many repetitions	reference distribution and uncertainty
Particle size	select largest particles first	tests contact-size advantage
Degree ranking	select particles with most contacts	local connectivity baseline
Heat-path ranking	rank particles by homogeneous heat throughput, $\sum_j G_{ij}(T_i - T_j) $	boundary-aware physics baseline
Fixed-budget swap optimization	exchange one selected and one unselected particle; retain improving swaps	locally optimized allocation with exactly M particles
Continuous relaxation	optimize soft material variables and project to exactly M particles	approximate upper-level optimizer

Performance is reported as absolute k_{eff} and as enhancement over the random mean,

$$\eta_{\text{alloc}} = \frac{k_{\text{eff}} - \bar{k}_{\text{eff,random}}}{\bar{k}_{\text{eff,random}}}. \quad (11)$$

To explain the improvement, the analysis will examine conductive-cluster spanning, directional tortuosity, fraction of the material budget in the main heat-carrying backbone, redundancy of parallel paths and the distribution of edge heat fluxes.

Controlled fixed-volume allocation pilot

A controlled contact-only allocation pilot was completed on `packing_seed_18427` with $k_L = 1 \text{ W m}^{-1} \text{ K}^{-1}$ and $k_H = 10 \text{ W m}^{-1} \text{ K}^{-1}$. Every heterogeneous assignment contains exactly 50 high-conductivity particles: 10 of the 100 particles with diameter 1.5 mm, 30 of the 300 particles with diameter 2.0 mm and 10 of the 100 particles with diameter 2.5 mm. The selected high-conductivity particle volume is therefore identical in every comparison, $V_H = 2.25147 \times 10^{-7} \text{ m}^3$. This removes the possibility that an allocation benefits simply by selecting more large particles.

The random reference comprises 100 independently sampled, quota-preserving assign-

ments. Degree and heat-path rankings are also applied separately within each size class. The swap search begins from the better transparent heuristic and exchanges selected and unselected particles only within the same size class. Five independent restarts were performed with 3000 proposed swaps per restart. Only swaps that increased k_{eff} were retained. Table 3 summarizes the controlled result.

Table 3: Controlled fixed-volume allocation result for `packing_seed_18427`. Enhancements are evaluated relative to the mean of 100 random allocations satisfying the 10/30/10 size-class quotas.

Allocation	k_{eff} [W/(m K)]	Enhancement over random mean
Random, mean $\pm$ standard deviation	0.11123 ± 0.00194	reference
Degree ranking	0.11362	2.1%
Heat-path ranking	0.12186	9.6%
Swap search, five-restart mean	0.17474 ± 0.00259	57.1%
Best allocation found	0.17895	60.9%

The homogeneous low-conductivity value is $0.09393 \text{ W m}^{-1} \text{ K}^{-1}$. The best allocation found is consequently 90.5% above the homogeneous bed, 52.6% above the best of the 100 random assignments and 46.8% above heat-path ranking. The random ensemble has a coefficient of variation of 1.75%, whereas the five final swap-search results have a mean of $0.17474 \pm 0.00259 \text{ W m}^{-1} \text{ K}^{-1}$ and a coefficient of variation of 1.48%. The small between-restart variation shows that the independent searches reach a similar high-performing region.

The best particle-ID assignment was subsequently evaluated in an independent call to the thermal solver. It reproduced $k_{\text{eff}} = 0.1789481826 \text{ W m}^{-1} \text{ K}^{-1}$ exactly, with 50 high-conductivity particles, the original 1343 solid-contact edges and a relative hot-cold energy-balance error of 2.93×10^{-12} . The graph retains one 497-particle boundary-spanning component and three isolated particles, confirming that the improvement results from the material labels rather than a change in packing topology.

Accepted improvements still occurred close to proposal 3000 in several restarts. The reported assignment is therefore not claimed to be a global or strict local optimum. Instead, the algorithm is defined as a reproducible fixed-budget stochastic swap search: 3000 proposals per restart and five restarts, with the best and mean performance both reported. Under this protocol, the large controlled improvement supports continuing the

study across the remaining independent packings.

WP4: Generalization, optimization quality and dataset expansion

For a 500-particle graph, exhaustive search over all $\binom{N}{M}$ allocations is impossible except for very small diagnostic graphs. Exact enumeration on small subgraphs and reduced packings is therefore used to measure the optimality gap of greedy and relaxation-based methods. On full packings, multiple optimizer starts and convergence histories provide a practical reference.

The ten current packings are first used as a method-development and variability pilot. If the ranking of allocation strategies is stable and the effect size is large, the physical optimization study can be expanded to approximately 30–50 independent packings for statistical assessment. If graph learning is pursued, a larger ensemble will normally be required; an initial target of 100–300 packings should be confirmed using learning curves rather than treated as a fixed requirement. A packing-level split is mandatory: all allocations derived from one DEM contact network must remain in the same training, validation or test partition.

Robustness tests should include unseen random realizations, alternative conductive fractions, conductivity contrasts and, later, a larger particle count. The primary claim should be based on transfer to unseen contact networks, not on predicting additional allocations on a packing already seen during training.

WP5: Conditional graph-learning placement policy

The GNN is not required to solve the thermal equations and is not initially a reinforcement-learning task. A simpler supervised formulation is preferred. Node features can include normalized coordinates, radius, coordination, distance from both thermal boundaries and local contact statistics. Edge features can include normalized centre distance, contact radius or conductance, overlap and orientation relative to the imposed temperature gradient. Global inputs include the conductive fraction and conductivity ratio.

The GNN produces one allocation score s_i per particle. The M particles with the highest scores are assigned the conductive material, enforcing the budget exactly. Training targets may be derived from high-quality greedy or relaxation-based assignments, while a differentiable physics loss can directly reward high k_{eff} if the solver is implemented in

a differentiable form. The learned policy is valuable only if it offers at least one of the following:

1. improved k_{eff} relative to strong transparent baselines;
2. comparable quality at substantially lower allocation time; or
3. transfer across unseen packings, system sizes or material budgets that fixed hand-designed rankings do not provide.

If these conditions are not met, the first paper should remain a DEM–network inverse-allocation study without claiming that a GNN is necessary.

4 Validation criteria and publication logic

The study is successful only if the following conditions are met:

1. all accepted DEM cases satisfy the prescribed PSD and porosity and have mechanically reasonable contact networks;
2. the thermal solver satisfies energy conservation and basic numerical invariance tests;
3. allocation produces a reproducible improvement over a sufficiently sampled random baseline across independent packings;
4. the improvement is robust to reasonable contact-conductance models and is explained through measurable heat-carrying network structure; and
5. any claimed GNN advantage is demonstrated on entirely unseen DEM packings against strong physics and graph baselines.

The minimum publishable contribution is the physical inverse-allocation study: a controlled DEM ensemble, verified network solver, fixed-budget optimization, strong baselines and interpretable topological design rules. A second or expanded contribution is a transferable GNN allocation policy. This separation limits project risk: failure of the GNN to outperform a greedy physical method does not invalidate the thermal-design question.

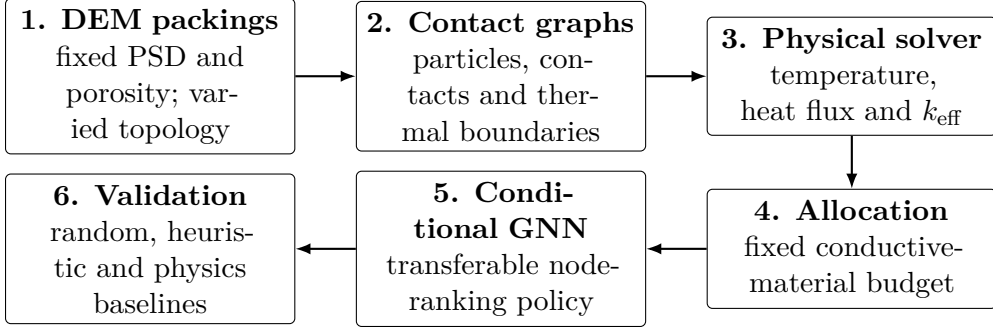


Figure 2: Proposed workflow. DEM is run once per packing; the thermal solver can evaluate many material allocations on the same fixed contact graph.

5 Expected outcome and relevance

The expected first outcome is a quantitative map of when conductive-particle placement matters as a function of material fraction, conductivity contrast, the competition between solid and stagnant-fluid conduction, and contact- network realization. A useful organizing quantity is the ratio of a characteristic contact conductance to a characteristic fluid-gap conductance. This separates fluid-dominated, mixed-conduction and solid-network-dominated regimes. The second outcome is an interpretable description of effective allocations in terms of hot-cold spanning paths, bottlenecks, parallel-path redundancy and particle/contact size. The third, conditional outcome is a graph model that recommends an allocation on an unseen packing without performing a long combinatorial search.

The framework is relevant to granular thermal-energy-storage media, thermally enhanced packed beds, composite particulate materials and other systems in which a high-conductivity additive is limited by cost, mass or processability. Application-specific claims should be made only after selecting realistic material pairs and including the dominant additional heat-transfer mechanisms for that application.

6 Principal limitations and decision points

The intended model includes stagnant-fluid conduction but not forced convection. This is a deliberate physical scope appropriate to static beds with negligible fluid motion; it is not intended to represent actively ventilated or flowing packed-bed reactors. Natural convection must be excluded through the pore-scale Rayleigh-number check. Radiation remains negligible only at moderate temperatures and should be added for high-temperature

applications. Second, DEM overlap is a numerical and mechanical quantity; using it directly to calculate contact area couples the thermal result to stiffness and confinement assumptions. Sensitivity to the conductance law is therefore mandatory. Third, prescribing the number of high-conductivity particles is not identical to prescribing their mass or cost when the bed is polydisperse. The final formulation should compare number-, volume- and cost- constrained budgets where appropriate. Fourth, ten packings are adequate for workflow development but not necessarily for training a transferable GNN. Finally, numerical optimization produces a theoretical material assignment; manufacturing or mixing constraints must be added before claiming direct experimental realizability.

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